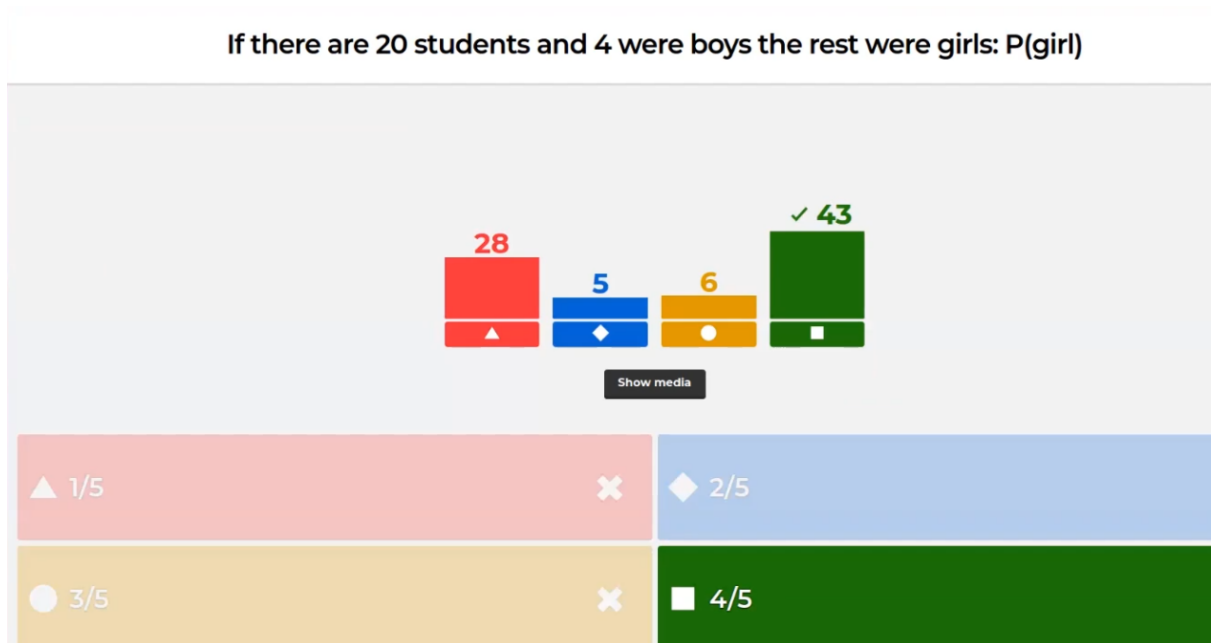


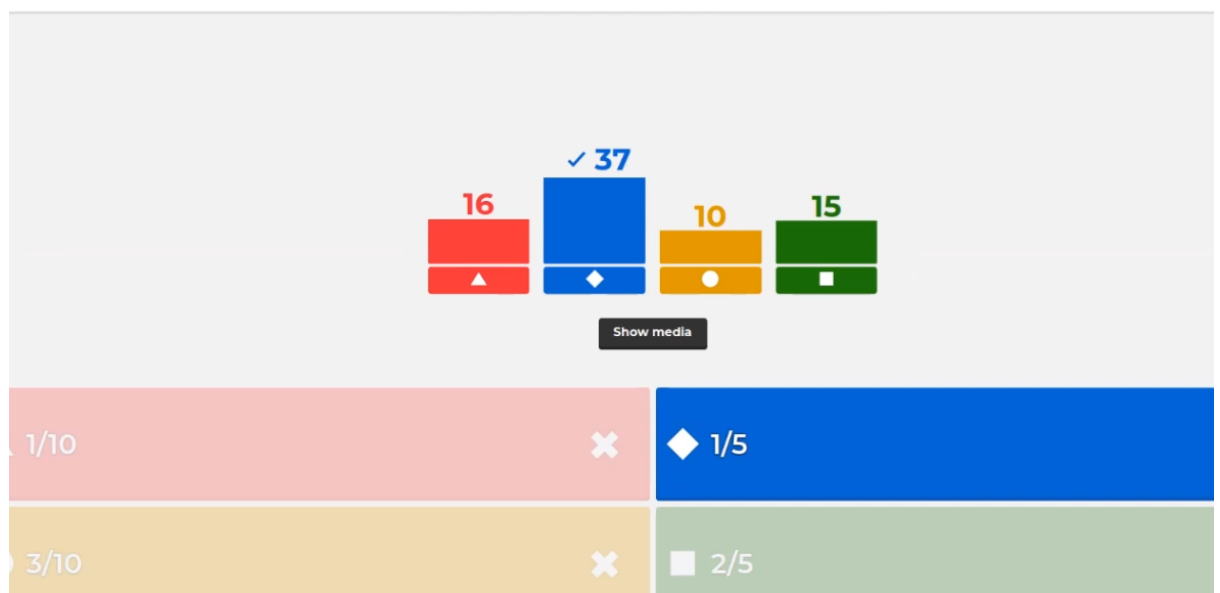
Cursul 2

1. If there are 20 students and 4 were boys the rest were girls: $P(\text{girl})$



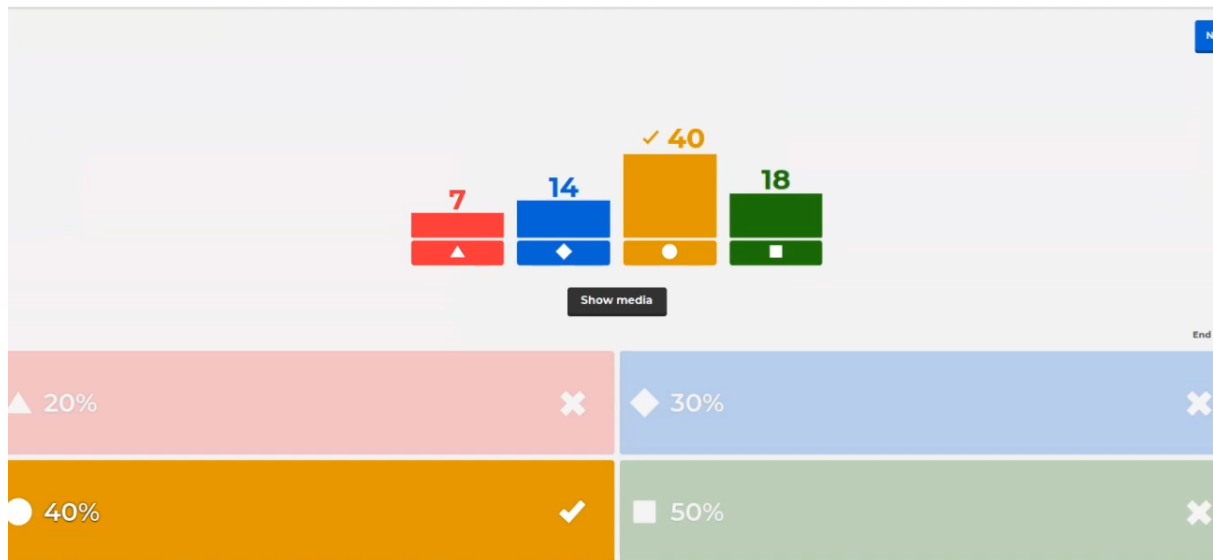
2. What is the experimental probability of getting a 2 for the 50 rolls of the die below?

What is the experimental probability of getting a 2 for the 50 rolls of the die below?



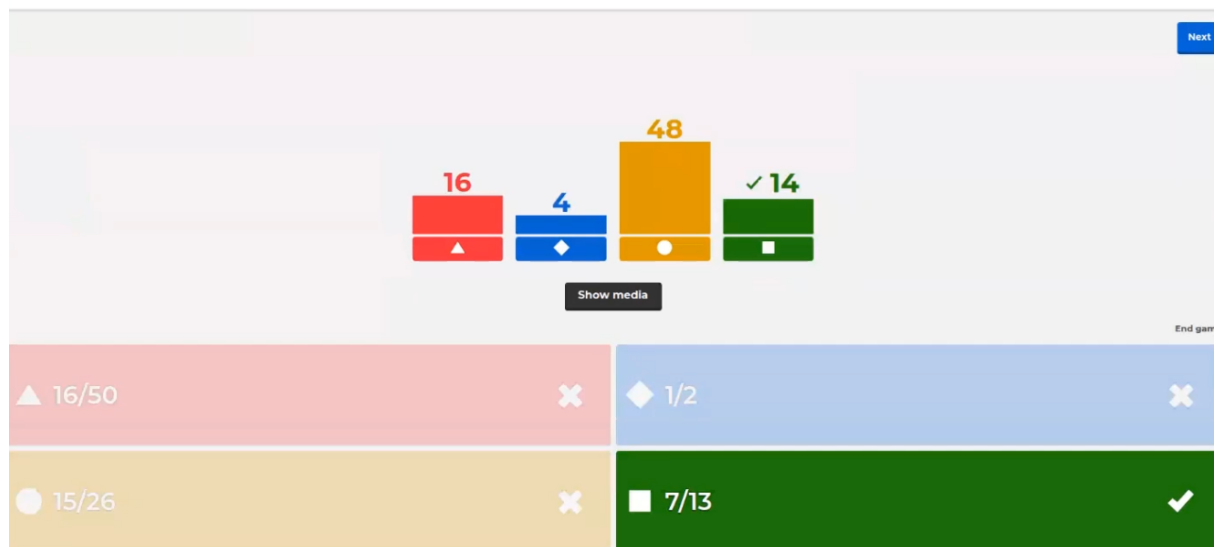
3. What is the theoretical probability for getting a blue or red marble?

What is the theoretical probability for getting a blue or red marble?



4. What is the probability of getting a red card or a king in a deck of cards?

What is the probability of getting a red card or a king in a deck of cards?



5. What is the probability of picking a vowel in the word: DICTIONARY

What is the probability of picking a vowel in the word: DICTIONARY

Next

7 5 4 ✓ 7

▲ 6/10 ✕ ◆ 3/10 ✕ ● 3/5 ✕ ■ 2/5 ✓

Show media

End game

6. What is the probability of getting a multiple of 3 on a die and tails on a coin?

What is the probability of getting a multiple of 3 on a die and tails on a coin?

Next

✓ 29 22 9 20

▲ 1/6 ✓ ◆ 2/6 ✕ ● 1/4 ✕ ■ 33% ✕

Show media

End game

Cursul 3

- Which is not an exact inference algorithm?

Which is not an exact inference algorithm?

16 6 23 20

▲ ◆ ● ■

Show media

▲ Inference by enumeration	✗	◆ Inference by variable enumeration	✗
● Inference by stochastic simulation	✓	■ Inference by Markov Chain Monte Carlo	✓

End game

- Which are hidden variables for the query $P(A_{j,m})$ in the alarm network?

Which are hidden variables for the query $P(A_{j,m})$ in the alarm network?

24 32 6 3

▲ ◆ ● ■

Show media

▲ J,M	✗	◆ B,E	✓
● A	✗	■ J,A,M	✗

End game

3. Which are evidence variables for the query $P(A_{j,m})$ in the alarm network?

Which are evidence variables for the query $P(A_{j,m})$ in the alarm network?

Next

✓ 46

1 8 9

Show media

End game

▲ J,M	✓	◆ B,E	✗
● A	✗	■ J,A,M	✗

4. Factors are used in which algorithm?

Factors are used in which algorithm?

Next

✓ 35

11 14 5

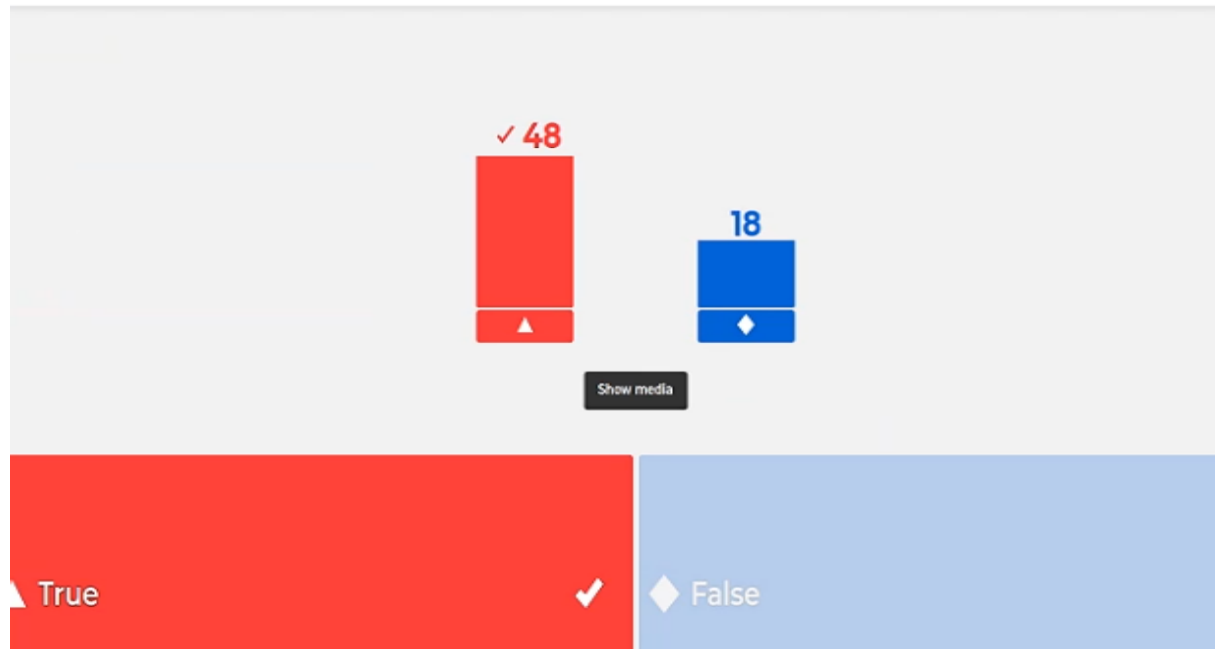
Show media

End game

▲ Inference by enumeration	✗	◆ Inference by variable elimination	✓
● Inference by stochastic simulation	✗	■ Inference by Markov Chain Monte Carlo	✗

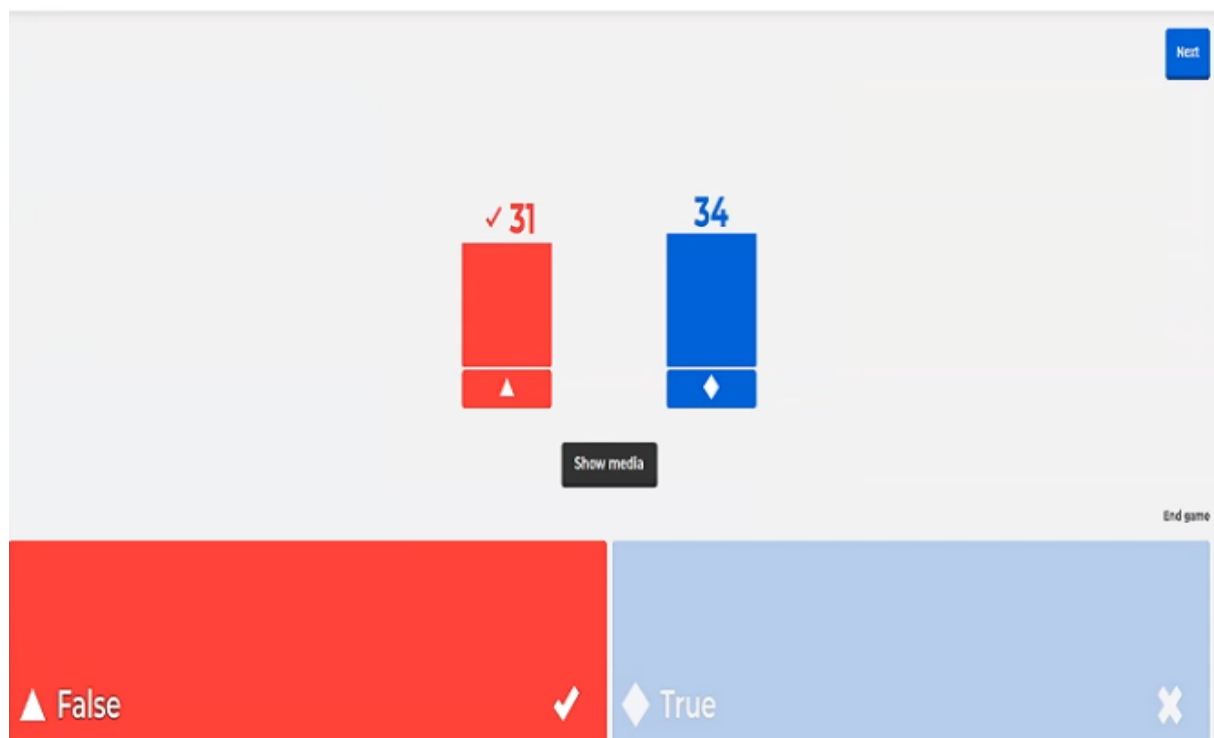
5. Rejection sampling generates only events that are consistent with the evidence e

Rejection sampling generates only events that are consistent with the evidence e



6. MCMC generates samples from scratch

MCMC generates samples from scratch



7. Likelihood weighting

Likelihood weighting

✓ 43 9 13

▲ ◆ ●

Show media

Next

End game

▲ uses evidence to weight samples ✓	◆ samples from an empty network ✗
● rejects samples disagreeing with evidence ✗	

Curs 4

1. Markov assumption - current state depends on

Markov assumption - current state depends on

27 ✓ 24 19

▲ ◆ ●

Show media

Next

End game

▲ the previous state ✗	◆ only a finite fixed number of previous states ✓
● its parents + children + children's parents ✗	

2. A stationary process

A stationary process

✓ 38 23 9

▲ ◆ ●

Show media

End game

▲ a process of change, but the laws of change don't change ✓

◆ the process is static (the state does not change) ✗

● none of the above ✗

3. Sensor Markov assumption

Sensor Markov assumption:

14 7 ✓ 25 23

▲ ◆ ● ■

Show media

End game

▲ $P(E_t | X_{0:t}, E_{0:t-1}) = P(E_t | X_{t-1}, E_{0:t-1})$ ✗

◆ $P(E_t | X_{0:t}, E_{0:t-1}) = P(E_t | X_{0:t})$ ✗

● $P(E_t | X_{0:t}, E_{0:t-1}) = P(E_t | \text{■})$ ✓

■ $P(E_t | X_{0:t}, E_{0:t-1}) = P(E_t | X_{t-1})$ ✗

4. Filtering (state estimation) is

Filtering (state estimation) is

$\triangle P(X_1 e_{1:t})$ ✓	$\diamond P(X_{t+k} e_{1:t})$ ✗
$\bullet P(X_k e_{1:t}), k < t$ ✗	$\blacksquare \operatorname{argmax}(X_{1:t}) P(X_{1:t} e_{1:t})$ ✗

End game

5. Smoothing is

Smoothing is

$\triangle P(X_1 e_{1:t})$ ✗	$\diamond P(X_{t+k} e_{1:t})$ ✗
$\bullet P(X_k e_{1:t}), k \leq t$ ✓	$\blacksquare \operatorname{argmax}(X_{1:t}) P(X_{1:t} e_{1:t})$ ✗

End game

6. Prediction is

Prediction is

4 35 2 27

▲ ◆ ● ■

Show media

Next

End game

▲ $P(X_t e_{1:t})$	✗	◆ $P(X_{t+k} e_{1:t})$	✓
● $P(X_k e_{1:t}), k < t$	✗	■ $\text{argmax}_{(X_{1:t})} P(X_{1:t} e_{1:t})$	✗

7. Finding the most likely explanation

Finding the most likely explanation

14 10 6 38

▲ ◆ ● ■

Show media

Next

End game

▲ $\text{argmax}_{(X_{1:t})} P(X_{1:t} e_{1:t})$	✗	◆ $\text{argmax}_{(X_{1:t})} P(X_k e_{1:t}), k < t$	✗
● $\text{argmax}_{(X_{1:t})} P(X_{1:t} e_t)$	✗	■ $\text{argmax}_{(X_{1:t})} P(X_{1:t} e_{1:t})$	✓

8. Which is not an inference task in a temporal model?

Which is not an inference task in a temporal model?

33 4 12 18

▲ ◆ ● ■

Show media

Next

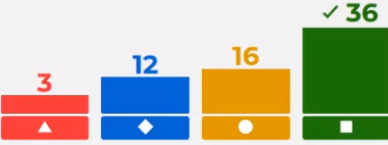
End game

▲ Resolution	✓	◆ Prediction	✗
● Smoothing	✗	■ Finding the most likely explanation	✗

9. The reason in dynamic bayesian network you do not need

To reason in a dynamic bayesian network you do not need

Next



Show media

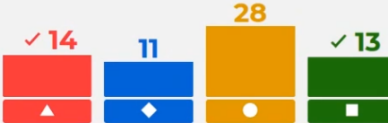
End game

▲ initial state model	✗	◆ transition model	✗
● sensor model	✗	■ gaussian distribution	✓

10. Learning of dynamic bayesian network means to

Learning of dynamic bayesian network means to

Next



Show media

End game

▲ learn from observations the transition model	✓	◆ interpret each node in the network as a neuron	✗
● smoothing, prediction, state estimation	✗	■ learn from observations the sensor model	✓

Curs 5

1. Which is not true about Kalman filtering

Which is not true about Kalman filtering

Next

17 34 11

Show media

End game

▲ used for state estimation from noisy observations over time ✗

◆ handles discrete variables ✓

● the noise is gaussian ✗

2. Which is true about basic Kalman filter?

Which is true about basic Kalman filter?

Next

20 39

Show media

End game

▲ it handles non gaussian distributions ✗

◆ cannot be applied if transition model is nonlinear ✓

3. Kalman filter cannot be applied to

Kalman filter cannot be applied to

Next

7 14 12 29

Show media

End game

▲ radar tracking of aircraft and missile ✗

◆ visual tracking of vehicles and people ✗

● ocean current from satellite surface measurements ✗

■ current location of the lost keys ✓

4. To construct a Dynamic Bayesian Network you do not need
To construct a Dynamic Bayesian Network you do not need

Next

Show media

End game

▲ the prior distribution over the state variables $P(X_0)$	✗	◆ the transition model $P(X_{t+1} X_t)$	✗
● full joint probability table	✓	■ the sensor model $P(E_t X_t)$	✗

5. In case of transient failure
In case of transient failure

Next

Show media

End game

▲ sensor sends a message of type "I am broken"	✗	◆ sensor occasionally sends some nonsense	✓
--	---	---	---

6. The persistent failure model
The persistent failure model

Next

Show media

End game

▲ describes how the sensor behave in case of failure	✓	◆ says that the sensor is broken	✗
--	---	----------------------------------	---

7. The probability $P(B_{t+1}|B_t)=1$ in the CTP of the persistence arc says that

8. Which is not true about particle filtering?

Which is not true about particle filtering?

Next

3 15 ✓ 12

Show media

End game

▲ each sample is propagated based on the transition model ✕

◆ initial samples are generated based on prior distribution ✕

● the number of samples (N) may change during filtering ✓

9. Which is not true about particle filtering?

Which is not true about particle filtering?

Skip

11

13 Answers

▲ sample is weighted by the likelihood it assigns to evidence

◆ is an exact inference method for DBNs

● sample is weighted by the sensor model $P(e_{t+1}|x_{t+1})$

Curs 7 - la mijloc

1. A micromort is

A micromort is

15

▲

✓ 19

◆

0

●

[Show media](#)

End game

▲ the unit measure of utility

✕

◆ one milionth chance of deadh

✓

● quality adjusted life years

✕

2. The basis of insurance premium is that people are

The basis of insurance premium is that people are

6

▲

✓ 22

◆

7

●

[Show media](#)

End game

▲ risk-neutral

✕

◆ risk-averse

✓

● risk-seeking

✕

3. Why doesn't money behave us a utility function?

Why doesn't money behave as a utility function?

14

▲

4

◆

✓ 9

●

8

■

[Show media](#)

End game

▲ Money is its own value

✕

◆ People are risk-prone

✕

● People are risk-averse

✓

■ You can't put a value on money

✕

4. What is the MEU principle used for?

What is the MEU principle used for?

Next

23

▲

5

◆

✓ 4

●

3

■

Show media

End game

▲ Minimize Expected Utility	✗	◆ Calculate utility	✗
● Action selection	✓	■ Explain irrational behaviour	✗

5. What is the points of the agents in this course?

What is the points of the agents in this course?

Next

4

▲

✓ 21

◆

4

●

4

■

Show media

End game

▲ Acting deterministically	✗	◆ Acting rational	✓
● Acting human	✗	■ Acting quickly	✗

6. What do you introduce to a bayesian network to make it a decision network?

What do you introduce to a bayesian network to make it a decision network?

Next

✓ 19

▲

11

◆

0

●

3

■

Show media

End game

▲ Action and utility nodes	✓	◆ Utility nodes	✗
● MEU node	✗	■ Nothing, it is already a decision network	✗

7. The expected value of information (VPI) is

The expected value of information (VPI) is

3 8 13 ✓ 8

▲ ◆ ● ■

Show media

Next

End game

▲ negative	✕	◆ additive	✕
● order-dependent	✕	■ non of the above	✓

8. cognitive bias where one relies on an initial piece of information to make decisions
cognitive bias where one relies on an initial piece of information to make decisions

1 8 3 ✓ 20

▲ ◆ ● ■

Show media

Next

End game

▲ risk averse	✕	◆ certainty effect	✕
● framing	✕	■ anchor effect	✓

9. Cognitive bias where one decides based on if the options are presented positively or negatively

Cognitive bias where one decides based on if the options are presented positively or negatively

8 11 ✓ 12 1

▲ ◆ ● ■

Show media

Next

End game

▲ risk averse	✕	◆ certainty effect	✕
● framing	✓	■ anchor effect	✕

10. Theory that describes how a rational agent should act

Theory that describes how a rational agent should act.

A Kahoot! quiz slide with a light gray background. At the top right is a 'Next' button. In the center, there are two podiums: a red one on the left with a score of 12 and a blue one on the right with a score of 20. Below the podiums is a 'Show media' button. At the bottom, there are two answer boxes: a red one on the left and a blue one on the right. The red box contains the text '▲ Normative theory' with a checkmark icon. The blue box contains the text '◆ Descriptive theory' with an 'X' icon. The text 'End game' is visible in the bottom right corner.

Curs 8- kahoot de inceput

1. Sequential decision problems (SDP)

Sequential decision problems (SDP)

A Kahoot! quiz slide with a light gray background. At the top right is a 'Next' button. In the center, there are four podiums: red (4), blue (24), yellow (1), and green (11). Below the podiums is a 'Show media' button. At the bottom, there are four answer boxes: red, blue, yellow, and green. The red box contains '▲ incorporate utility, uncertainty and sensing' with a checkmark. The blue box contains '◆ agent's utility depends on a sequence of decisions' with a checkmark. The yellow box contains '● are a special case of planning' with an 'X'. The green box contains '■ are a special case of Markov Decision Problems' with an 'X'. The text 'End game' is visible in the bottom right corner.

2. A Markovian decision process is a sequential decision problem for environments

A Markovian decision process is a sequential decision problem for environments

A Kahoot! quiz slide with a light gray background. At the top right is a 'Next' button. In the center, there are four podiums: red (4), blue (7), yellow (21), and green (8). Below the podiums is a 'Show media' button. At the bottom, there are four answer boxes: red, blue, yellow, and green. The red box contains '▲ fully observable and stochastic' with a checkmark. The blue box contains '◆ fully observable, and deterministic' with an 'X'. The yellow box contains '● with Markov transition models and additive rewards' with a checkmark. The green box contains '■ partial observable and stochastic' with an 'X'. The text 'End game' is visible in the bottom right corner.

3. An optimal policy $\pi(s)$

An optimal policy $\pi(s)$

9

✓ 21

✓ 10

Show media

End game

▲ minimises the expected sum of rewards

✗

◆ specifies the best action for every possible state s

✓

● yields the highest expected utility

✓

4. An optimal policy for a finite horizon is

An optimal policy for a **finite horizon** is

An optimal policy for a **finite horizon is**

✓ 15

25

Show media

End game

▲ nonstationary

✓

◆ stationary

✗

5. When the discount factor is close to 0, rewards in the distance future are viewed as

When the discount factor is close to 0, rewards in the distant future are viewed as

When the discount factor is close to 0, rewards in the distant future are viewed as

✓ 19

21

Show media

End game

▲ insignificant

✓

◆ significant

✗

6. Bellman equation

Bellman equation

6 7 17 9

▲ ◆ ● ■

Show media

End game

▲ $U(s) = R(s) + \gamma \max_{a \in A(s)} \sum_{s'} P(s' a, s) U(s')$	✗	◆ $U(s) = R(s) + \max_{a \in A(s)} \sum_{s'} P(s' a, s) U(s')$	✗
● $U(s) = R(s) + \gamma \max_{a \in A(s)} \sum_{s'} P(s' a, s) U(s')$	✓	■ $U(s) = R(s) + \gamma \max_{a \in A(s)} \sum_{s'} P(s' a, s) U(s')$	✗

7. An outcome is Pareto optimal if

An outcome is Pareto optimal if

12 19 1 7

▲ ◆ ● ■

Show media

End game

▲ there is no other outcome that one players would prefer	✗	◆ there is no other outcome that all players would prefer	✓
● there is other outcome that one players would prefer	✗	■ there is other outcome that all players would prefer	✗

8. Which of the following auctions are public and descending

Which of the following auctions are public and descending

9 9 11 9

▲ ◆ ● ■

Show media

End game

▲ English	✗	◆ Dutch	✓
● Sealed bid	✗	■ Vickrey	✗

Curs 8 Kahoot de final

1. When the output y (goal predicate) has a finite set of values, the learning problem is

When the output y (goal predicate) has a finite set of values, the learning problem is called

20

✓ 21

Show media

End game

▲ regression ✗

◆ classification ✓

2. The principle "Ockham's razor" prefers

The principle "Ockham's razor" prefers

17

✓ 23

1

Show media

End game

▲ the best hypothesis consistent with data ✗

◆ prefers the simplest hypothesis consistent with data ✓

● $X^2+1/3$ over X^3 ✗

3. The learning problem is realizable if

The learning problem is realizable if

✓ 19

13

9

Show media

End game

▲ the hypothesis space contains the true function ✓

◆ if the data is consistent and without noise ✗

● the adequate ML algorithm is applied ✗

4. To select the next attribute, decision tree learning adopts

To select the next attribute, decision tree learning adopts

Next

End game

▲ a deep-first strategy	✗	◆ breadth first strategy	✗
● a greedy divide-and conquer strategy	✓	■ a randomised strategy	✗

5. If there is no attribute left, but both positive and negative examples, it means that

If there are no attributes left , but both positive and negative examples, it means that

Next

End game

▲ there is an error or noise in data	✓	◆ the domain is nondeterministic	✓
● can't observe an attribute that would distinguish examples	✓	■ none of the above	✗

6. More information means

More information means

Next

End game

▲ more entropy	✗	◆ less entropy	✓
----------------	---	----------------	---

7. Overfitting is

Overfitting is

Next

17

✓ 23

Show media

End game

▲ good ✕

◆ bad ✓

8. Overfitting

Overfitting

Next

✓ 5

✓ 10

✓ 5

✓ 20

Show media

End game

▲ violates Occam's razor ✓

◆ means overtraining ✓

● includes more parameters than needed ✓

■ all of the above ✓

9. Hold-out cross validation fails to use all available data

Hold-out cross validation fails to use all available data

Hold-out cross validation fails to use all available data

Next

6

✓ 16

18

Show media

End game

▲ False ✕

◆ True ✓

● De verificat!!!!!!! ✕

10. Peeking occurs when

Peeking occurs when

✓ 15 ✓ 13 8 4

▲ ◆ ● ■

Show media

Next

End game

▲ hypothesis was selected based on its test set error rate ✓	◆ no validation test was used ✓
● the learning curve is a happy graph ✗	■ the goal attribute is continuous ✗

Curs11(kahoot) 10.05.2021

1. Is confidence a symmetric measure

✓ 22 15 3

▲ ◆ ●

Show media

Next

End game

▲ No ✓	◆ Yes ✗
● Maybe ✗	

2. How many rules can be generated with d unique items

How many rules can be generated with d unique items?

Next

10 ▲	17 ◆	✓ 10 ●	6 ■
---------	---------	-----------	--------

Show media

End game

▲ $3^{d+1} - 2^d + 1$ ✗	◆ $3^{d+1} + 2^d + 1$ ✗
● $3^d - 2^{d+1} + 1$ ✓	■ $3^d + 2^{d+1} + 1$ ✗

3. Given d items how many possible candidate itemsets are there?

Given d items how many possible candidate itemsets are there?

Next

✓ 20 ▲	2 ◆	9 ●	11 ■
-----------	--------	--------	---------

Show media

End game

▲ 2^d ✓	◆ d^2 ✗
● $d!$ ✗	■ $(d-1)!$ ✗

4. Support of an itemset does not exceed the support of its subsets

Support of an itemset does not exceed the support of its subsets

Next

12 ✓ 23 7

Show media

End game

▲ False ✕

◆ True ✓

● It depends on the itemset ✕

5. $X, Y, X \text{ INCLUS IN } Y, S(X) \leq S(Y)$

$\forall X, Y \quad X \subseteq Y \rightarrow s(X) \leq s(Y)$

Next

30 ✓ 12

Show media

End game

▲ True ✕

◆ False ✓

6. Given a frequent itemset L with ... , how many candidate association rules can be generated?

Given a frequent itemset L with $|L|=k$, how many candidate association rules can be generated?

Next

11

▲

12

◆

✓ 16

●

3

■

Show media

End game

▲ 2^k ✕	◆ 2^{k-2} ✕
● 2^{k-2} ✓	■ $k!$ ✕

7. Confidence is anti-monotone w.r.t nr of items on the RHS

Confidence is anti-monotone w.r.t. nr of items on the RHS: $c(ABC \rightarrow D) \geq c(AB \rightarrow CD) \geq c(A \rightarrow BCD)$

Next

✓ 32

▲

10

◆

Show media

End game

▲ True ✓	◆ False ✕
---	--

8. Fk-1 x Fk-1 method. Which pair cannot be merged for F3

$F_{k-1} \times F_{k-1}$ method. Which pair cannot be merged for $F_3 = \{ABC, ABD, ABE, ACD, BCD, BDE, CDE\}$?

Next

6 ▲	10 ◆	✓ 13 ●	12 ■
--------	---------	-----------	---------

Show media

End game

▲ ABC, ABD	✗	◆ ABD, ABE	✗
● ABD, ACD	✓	■ ABC, ABE	✗

Curs 13

1. What is the main difference between REGRESSION and CLASSIFICATION?

What is the main difference between REGRESSION and CLASSIFICATION?

Next

✓ 24 ▲	3 ◆	4 ●	1 ■
-----------	--------	--------	--------

Show media

End game

▲ The nature - measurement of the target/labeled variable	✓	◆ The supervised / unsupervised nature of each one	✗
● The use of a statistical approach or not	✗	■ No difference	✗

2. If the hypothesis says it should be negative but in fact it is positive we have

If the hypothesis says it should be negative but in fact it is positive we have

Next

Show media

End game

▲ false positive	✗	◆ false negative	✓
● true positive	✗	■ true negative	✗

3. False positives are solved by

False positives are solved by

Next

Show media

End game

▲ generalization	✗	◆ specialization	✓
------------------	---	------------------	---

4. Generalization takes place by

Generalization takes place by

Next

Show media

End game

▲ adding conditions	✗	◆ dropping conditions	✓
---------------------	---	-----------------------	---

5. Which is not true about current-best - hypothesis search method?

Which is not true about current-best-hypothesis search method?



Next



Show media

End game

▲ maintains single hypothesis and adjusts it with new examples



◆ deterministic



● checks all previous examples for each modification



■ lot of backtracking

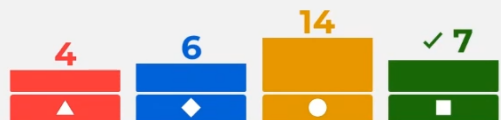


6. Which is not a stop condition for version spaces

Which is not a stop condition for version spaces



Next



Show media

End game

▲ one hypothesis left in the version space



◆ version space collapses (either S or G becomes empty)



● run out of examples and have several hypotheses remaining



■ the complexity of the hypothesis is over a threshold



7. FOIL algorithm avoids overfitting by

FOIL algorithm avoids overfitting by

Next

9 ✓ 10 13

Show media

End game

▲ performing cross-validation	✗	◆ penalising large hypothesis	✓
● using background knowledge	✗		

8. Relevance-based learning

Relevance-based learning

Next

20 ✓ 12

Show media

End game

▲ extracts general rules from single examples	✗	◆ uses prior knowledge in the form of determination	✓
---	---	---	---

9. Which is not true about version spaces?

Which is not true about version spaces?

Next

6 ✓ 6 12 8

Show media

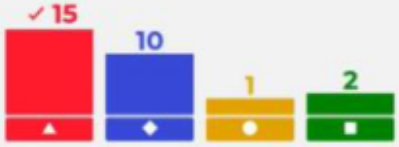
End game

▲ it is a hierarchical representation of knowledge	✗	◆ node values/attributes are continuous	✓
● is based on a generalization tree and a specialization tree	✗	■ it assumes data is correct	✗

Kahoot seria B

Given a Full Joint Prob. Distr., we can build a Bayes Net which can answer the same queries:

Given a Full Joint Prob. Distr., we can build a Bayes Net which can answer the same queries:



15 10 1 2

Show media

End game

▲ always ✓	◆ sometimes ✗
● never ✗	■ this cannot be proven ✗

Choose the BEST answer: the relation $P(a \vee b) = P(a) + P(b)$

Choose the BEST answer: the relation $P(a \vee b) = P(a) + P(b)$ is:



5 10 13 1

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End game

▲ always false ✗	◆ always true ✗
● sometimes true and sometimes false ✓	■ none of the above ✗

X and Y are independent. We have

X and Y are independent. We have: (a) $P(X \wedge Y) = P(X) \cdot P(Y)$ (b) $P(X|Y) = P(X)$ and (c) $P(Y|X) = P(Y)$

Next

6 4 3 11 ✓

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End game

▲ (a) and (b) are true, (c) is false	✗	◆ (a), (b) and (c) are false	✗
● (a) is true, (b) and (c) are false	✗	■ (a), (b) and (c) are true	✓

Choose the BEST answer: the relation $P(\neg a|B) = \dots$

Choose the BEST answer: the relation $P(\neg a|b) = 1 - P(a|b)$, where $P(\neg a)$ means $P(A = \text{False})$, is:

Next

4 11 ✓ 13 1

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End game

▲ always false	✗	◆ always true	✓
● sometimes false	✗	■ not provable	✗

Choose the BEST answer. Which of the following programs can play Go at top level:

Choose the BEST answer. Which of the following programs can play Go at top level:

Next

10 1 1 14 ✓

Show media

End game

▲ AlphaGo Master	✗	◆ AlphaGo Zero	✗
● AlphaZero	✗	■ all of the above	✓